

with sizes up to 1,024,000 atoms, measuring the performance over at least 5 ps of NPT dynamics at ambient pressure and 300 K. In all cases, the primary aim was to measure performance for a reasonably well-mixed system, and we did not attempt to reach full equilibration.

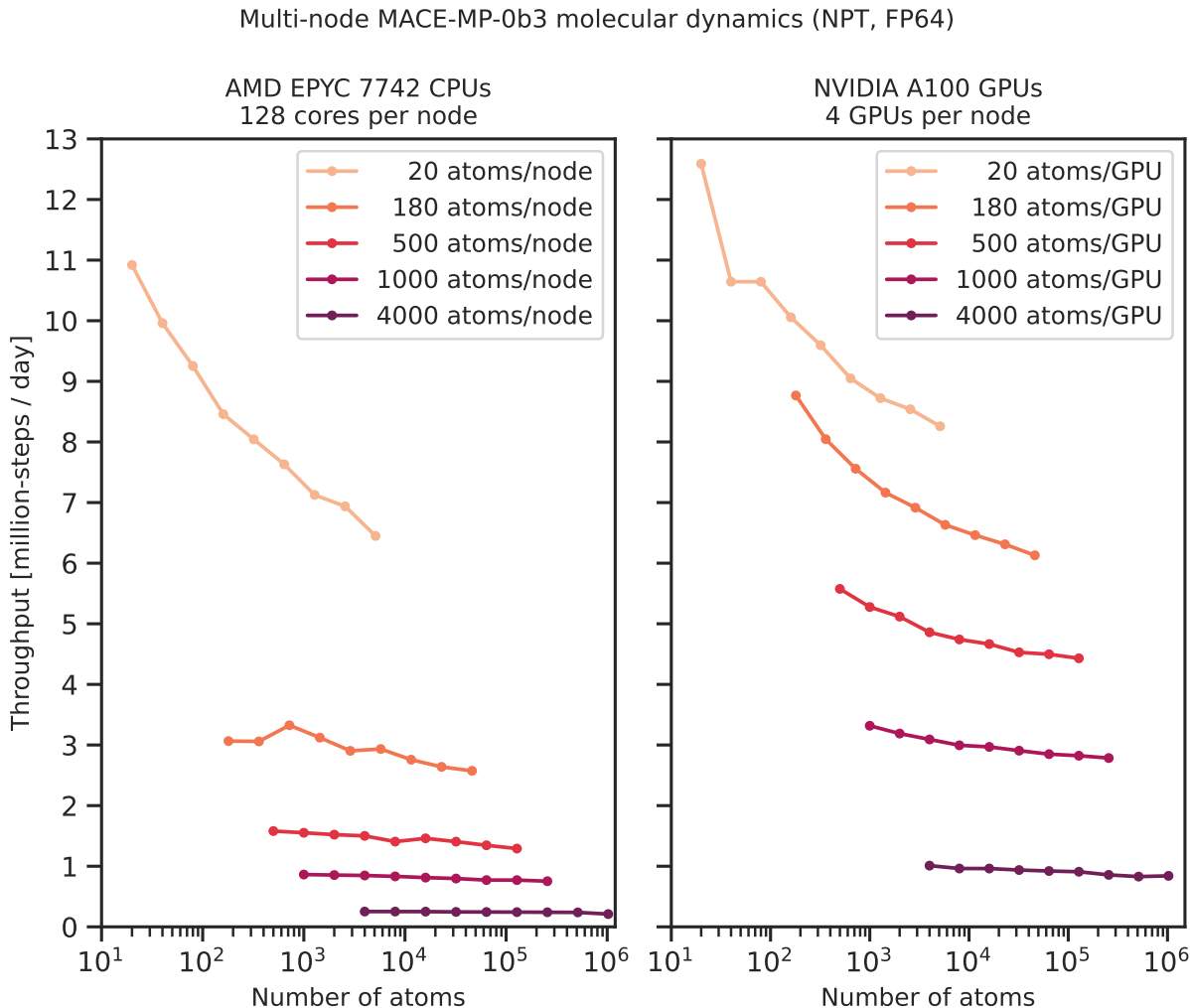


Figure S51: Double-precision multi-node performance for NPT dynamics in LAMMPS, measured for the 20-element Cantor alloy at ambient pressure and 300 K. The left plot contains data for 1-256 CPU nodes, with 128 cores per node, and the right plot has data for 1-256 A100 GPUs, with four GPUs per node. Near-perfect weak scaling is achieved with as little as a few thousand atoms per CPU node or GPU.

Similarity statement

Of the 150k MP structures, roughly 100k have at least one of the 20 elements used in this example, but only 3461 have compositions drawn exclusively from that set. Of this latter group, 193 are single-component structures, 2079 are binaries, 1155 are ternaries, and 34 are quaternaries. Moreover, no structure in the entire database has five or more of the 20 elements considered here. While it is encouraging that the dynamics appear stable for such a diverse, out-of-sample composition, we expect that incorporating dedicated datasets like that reported in (109) could enhance quantitative predictions.

A.33 MPA-0: Scaling up with the Alexandria dataset

To investigate the effect of training set size on model performance, we developed a MACE model, MPA-0, using identical hyperparameters to the MP-0b3 medium model but trained on an expanded dataset combining MPtraj and sAlex (304). The sAlex dataset comprises 10.5M structures extracted from the original Alexandria dataset, curated for compatibility with the Matbench Discovery benchmark. The Alexandria dataset consists only of randomly generated and relaxed crystal structures. Here, we demonstrate that increased data volume on just crystal structures significantly enhances model robustness across multiple applications. Notably, we observe strong enhancement in the accuracy of the MPA-0 model compared to MP-0b3 on applications for which coverage was not increased, namely amorphous phases and small molecules on surfaces.

A.33.1 Metals

Figure S52 compares generalized stacking fault profiles for body-centered cubic (BCC) metals (W, Mo, Nb) along (112)[111] and (110)[111] Γ -surfaces. MPA-0 shows markedly improved agreement with DFT reference data compared to MP-0b3 for W and Mo where MP-0b3 substantially underestimates the fault energies. For W and Mo, MPA-0 correctly captures the peak heights and the overall energy landscape, while maintaining proper symmetry across the normalized displacement. This improved accuracy in stacking fault energetics is critical for predicting mechanical properties and plastic deformation mechanisms in structural metals. This enhancement is likely due to an increase in chemical coverage for these metals in the Alexandria dataset.

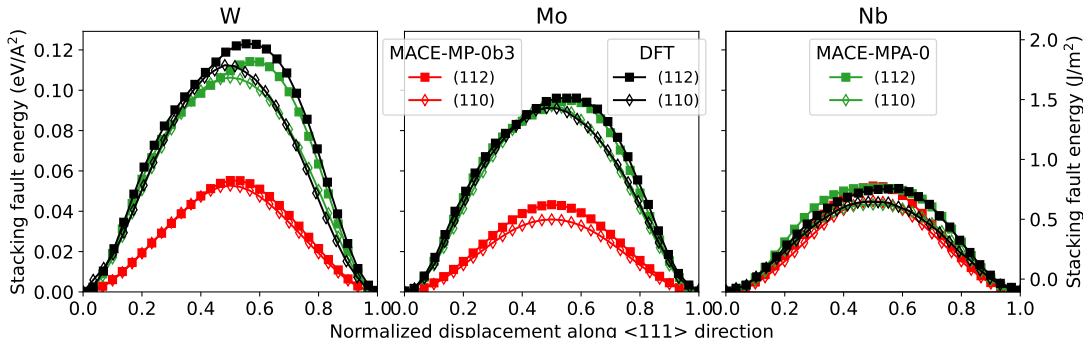


Figure S52: Generalized stacking fault profiles for (112)[111] and (110)[111] Γ -surfaces predicted by MP-0b3 (red) and MPA-0 (green) compared to DFT reference data (black). MPA-0 demonstrates improved agreement across all three BCC metals, especially for Nb.

A.33.2 Surfaces

The prediction of surface energetics represents a critical test for materials models. Figure S53 shows parity plots comparing the performance of MP-0b3+D3 and MPA-0+D3 in predicting relative energies between three structures across 157 molecule-surface combinations. MPA-0+D3 achieves a significantly lower MAD (0.28 eV vs. 0.38 eV), RMSD (0.37 eV vs. 0.52 eV), and higher Pearson correlation (0.92 vs. 0.86) compared to MP-0b3+D3. Fine-tuning further improves these metrics for MPA-0+D3 FT, reducing MAD to 0.25 eV and RMSD to 0.34 eV while increasing correlation to 0.94. This enhanced accuracy spans a diverse range of molecule-surface interactions involving 54 elements, indicating improved transferability for heterogeneous catalysis and materials interface applications. It is notable that Alexandria does not contain any surfaces, nor isolated small molecules, meaning that better coverage of ly crystal structureson improves even not directly related downstream tasks.

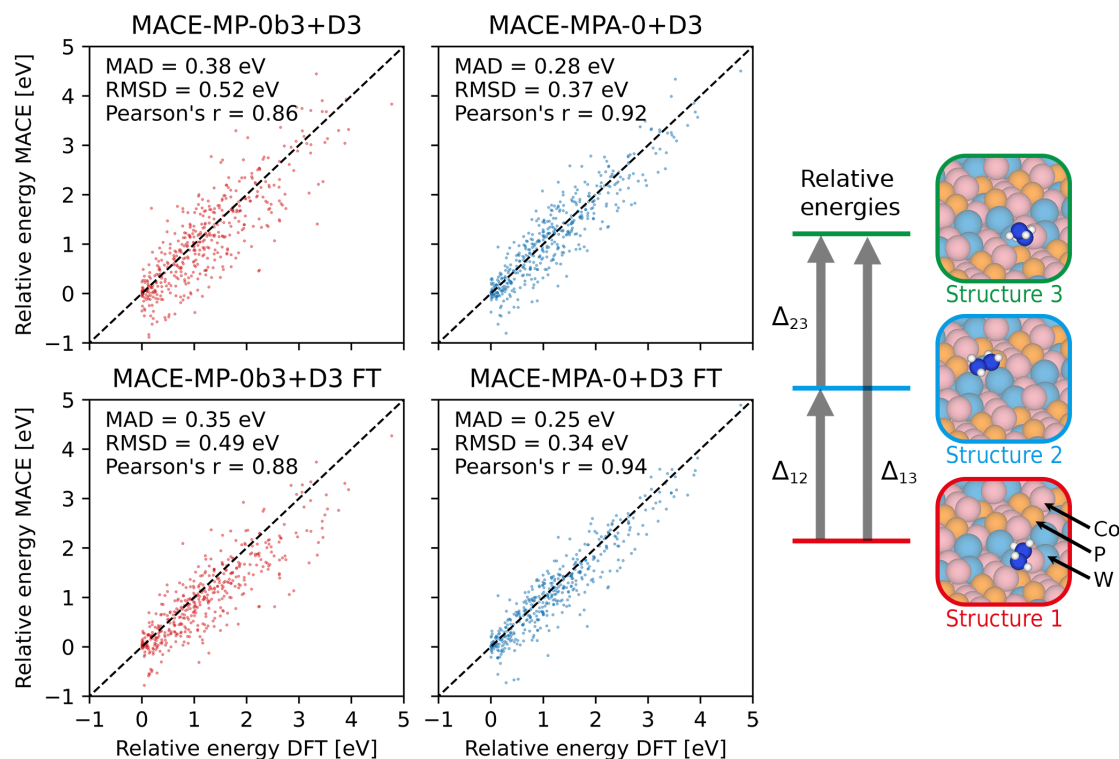


Figure S53: Comparison of MP-0b3+D3, MPA-0+D3 and their fine-tuned (FT) versions against DFT (PBE-D3(BJ)) in predicting relative energies between three structures across 157 molecule-surface combinations. MPA-0 demonstrates consistently smaller errors and improved correlation with DFT values.

A.33.3 Amorphous carbon

Accurate modeling of amorphous materials requires precise representation of local atomic environments and bonding. Figure S54 compares the structural properties of amorphous carbon predicted by MP-0b3, MPA-0, and the reference GAP-17 model. While MP-0b3 performs qualitatively well at low density (panel e-g), MPA-0 significantly improves the distribution of shortest-path ring sizes for the high density phases, showing a sharper peak at 6-membered rings (panel f) that closely aligns with GAP-17, whereas MP-0b3 displays a broader, less defined distribution across ring sizes 6-8. For coordination numbers at high density (panel h), MPA-0 reproduces the correct $sp^3:sp^2$ ratio with approximately 85% 4-coordinated and 15% 3-coordinated carbon atoms, closely matching GAP-17. In contrast, MP-0b3 shows excess 3-coordinated (sp^2) and 5-coordinated carbon atoms, indicating structural deficiencies in the predicted amorphous networks. The Alexandria dataset does not contain any amorphous structures, demonstrating that an increase in broad chemical coverage of crystals helps in predicting amorphous phases.

A.33.4 Metastable cell

A critical test for materials discovery applications is a model's ability to correctly rank metastable structures relative to the ground state. Figure S55 compares the error distributions of MP-0b3 and MPA-0 in predicting energy differences above the ground state ($\Delta E_{\text{MLIP}} - \Delta E_{\text{DFT}}$) across 16,126 relaxed structures from 100 different compositions. Both models exhibit a systematic tendency to overstabilize local minima, as indicated by negative errors that increase with distance from the ground state. However, MPA-0 shows a considerably narrower error distribution and significantly reduced overstabilization, particularly for structures with $\Delta E_{\text{DFT}} > 0.5$ eV/atom. This improved energy ranking enhances MPA-0's reliability for polymorph prediction, phase stability assessment, and materials discovery applications.

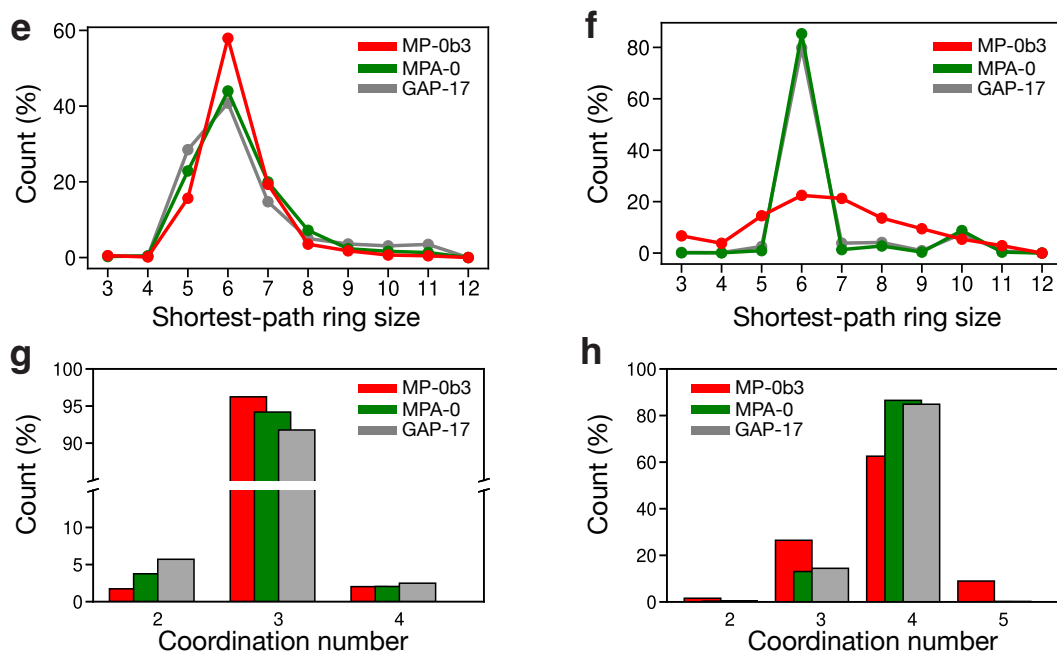


Figure S54: Structural analysis of 4,096-atom amorphous carbon configurations. (e-f) Shortest-path ring size distributions and (g-h) coordination number distributions for MP-0b3 (red), MPA-0 (green), and GAP-17 reference model (gray). MPA-0 better captures the correct bonding environments and medium-range order.

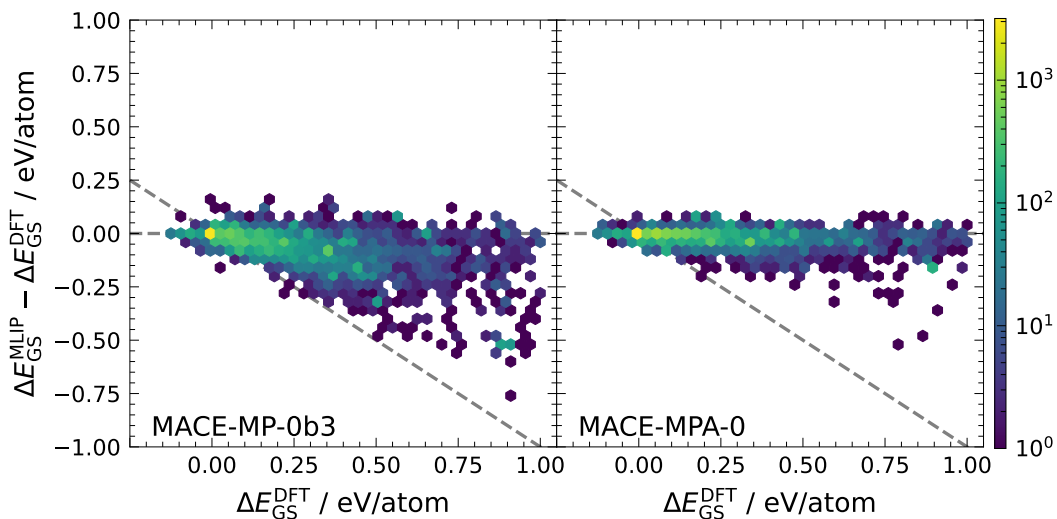
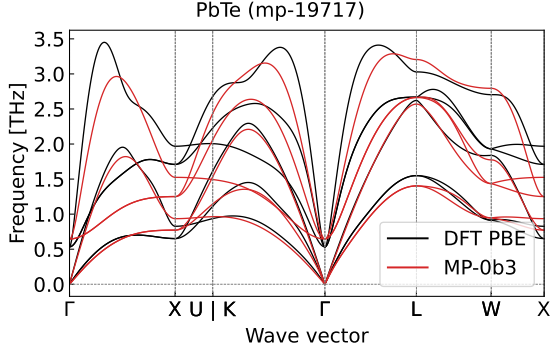


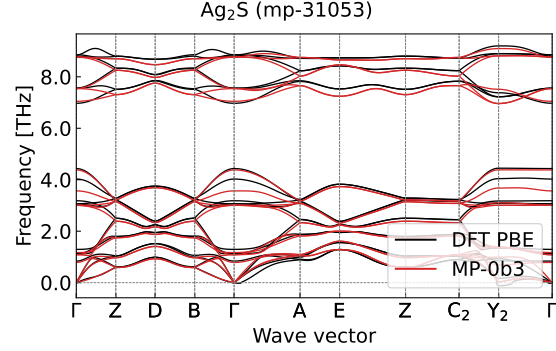
Figure S55: Errors in prediction of energies above the ground state for MP-0b3 (left) and MPA-0 (right) across 16,126 relaxed structures from 100 different compositions. The color scale indicates point density. Dashed lines represent zero error (horizontal) and the boundary where structures would be erroneously predicted to have lower energy than the true ground state (diagonal). MPA-0 shows reduced error magnitude and spread, particularly for higher-energy structures.

B Benchmarks

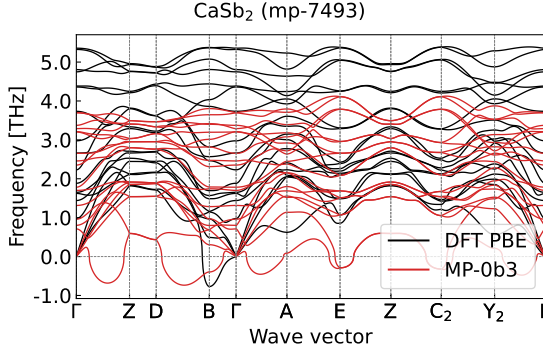
B.1 Phonons



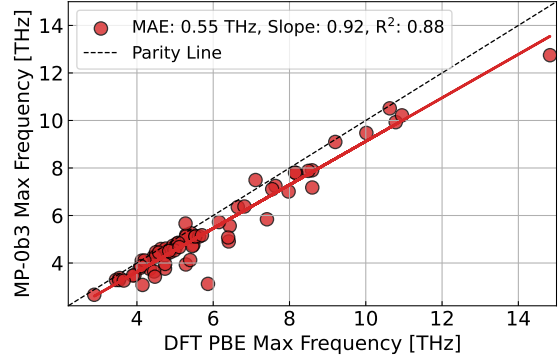
(a) PbTe phonon band structure



(b) Ag₂S phonon band structure



(c) CaSb₂ phonon band structure



(d) MACE vs PBE highest phonon frequency parity plot

Figure S56: Comparison of DFT PBE and MACE-MP-0b3 phonon band structures. Imaginary modes are plotted with a negative sign. a) and b) show examples of particularly good phonon bands. c) is a particularly bad example. d) Parity plot of MACE-MP-0b3 vs PBE highest phonon band frequency showing good agreement across the chosen materials. showing excellent agreement across diverse materials.

Accurate modeling of phonons is essential for determining dynamic stability of crystals, as well as thermoelectric properties and entropic contributions to the free energy (305, 306). Those are important factors in the discovery of new materials. Harmonic phonons are typically calculated from the restoring force on atomic displacements and require highly accurate force predictions to be physical. An ML potential trained on PBE forces should be able to reproduce PBE lattice vibrations (307, 308). To assess the accuracy of MACE-MP-0b3 restoring forces, we compare harmonic phonon modes predicted via the finite-displacement method as implemented in Phonopy (309, 310) and atomate2 (311, 312). For this purpose, we use a benchmark set of 70 thermoelectric, phase-change, and chalcogenide-based materials. Furthermore, the reference set only contains materials in which magnetism does not play a role and U-correction is not applied in the MP.

For all materials in the benchmark set we used the MPRelaxMaker (311) for structure optimization and chose a k-point density of 300 k-points per reciprocal volume and a cutoff energy of 520 eV (1.3*ENMAX of the highest ENMAX indicated in the pseudopotential files for the PAW method for each compound). We chose an electronic convergence criterion of 10^{-7} eV and the ionic convergence criterion of 10^{-6} eV. We perform the calculations of single-atom-displaced supercells with the MPStaticMaker (311), setting the